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EDITED BY

Patrick Ngulube,
University of South Africa, South Africa

REVIEWED BY

Diana Atuase,
University of Cape Coast, Ghana
Metwaly Eldakar,
Minia University, Egypt

*CORRESPONDENCE

Said Abubakar
✉ said@mu.edu.so

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AI use in academic research: Examining the mediating effect of user experience and the moderating effect of disciplinary context on perceived integrity

Said Abubakar* and Mohamed Yusuf Adan

Postgraduate Programme and Research, Mogadishu University, Mogadishu, Somalia

The study investigates academic researchers' use of AI in higher education research, drawing on the Unified Theory of Acceptance and Use of Technology (UTAUT) and integrity-trust frameworks. It examines the mediating effect of user experience and the moderating effect of disciplinary context on perceived integrity. A quantitative cross-sectional design was employed, collecting data via questionnaire from a sample of 100 academic researchers and lecturers across universities in Mogadishu, Somalia. Data analysis using SPSS and JAMOV (PROCESS macro) involved correlation, mediation (bootstrapping), and moderation analyses. The overall trend in the findings suggests a positive, and growing, acceptance of AI for academic research as researchers believe it helps them produce better research outcomes. User experience mediates significantly the relationship between AI use and perceived integrity but disciplinary context does not moderate this relationship. Though perceived integrity is generally high, the researchers have some concerns about the reliability and ethical use of AI in research.

KEYWORDS

academic research, AI use, mediating user experience, moderating disciplinary context, perceived integrity

Introduction

The rapid development of artificial intelligence (AI) technology has presented both new opportunities and greater ethical concerns in the context of academic research. AI systems are now routinely employed in data analytics, manuscript composition, and even peer-review, as well as in other processes where greater operational efficiency is desired (Lund and Naheem, 2023; Leung et al., 2023). This assimilation, however, brings with it certain challenges in regard to authorship, transparency, and academic fraud.

On the other hand, AI-assisted processes such as data analysis and drafting of manuscripts can augment the quality of research outputs. For example, AI assistance in manuscript drafting and editing has reduced literature synthesis times significantly and aids researchers in gap identification as well (Lund and Naheem, 2023; Leung et al., 2023). Owing to these undeniable advantages, many journals and academic institutions are now framing policies to define the involvement of AI in authorship, manuscript production and editing, and peer review (Lund and Naheem, 2023; Leung et al., 2023). These policies are not only helpful in resolving questions of contribution recognition, but also in protecting the scientific communication system from abuse (Lund and Naheem, 2023).

Conversely, integrating AI in research also generates substantial research ethics and research integrity issues. Different research publications have documented instances of

AI-created work resulting in duplication of effort, reinforcement of bias, and even creation of faked yet true-to-form manuscript (Chen et al., 2024; Májovský et al., 2023). The risk of these practices undermines the integrity of scientific journals and requires caution in approaches to using AI tools in disseminating research (Chen et al., 2024; Májovský et al., 2023). These issues call for rigorous and regular policies in academic journals to create standards for the amount of acceptable AI assistance and avoid such unethical practices of ghost writing or surreptitious data manipulation (Lund and Naheem, 2023; Guleria et al., 2023; Khlaif et al., 2023).

To address the positive and negative aspects of the function of AI in science, a set of scholars has placed some guidelines and ethical pointers on the use of this technology. Their initiatives range from the policies AI's contributions disregard and acknowledgment to the set high terms and requirements for the transparency, accountability, and validation of AI-generated data (Lund and Naheem, 2023; Leung et al., 2023; Hryciw et al., 2023). These actions among other productivity are the participation of traditional researchers and journal editors in the construction of guidelines for the responsible and ethical use of AI that promotes advancement without contravening the ethical principles of the university (Leung et al., 2023; Hryciw et al., 2023). Similarly, the authors also promise that by frequently updating these guidelines as the technology keeps evolving, they will be better in a position to say cheese to the emerging opportunities and at the same time scatter the possible harms related to meandering data production, thus achieving the replication barriers across a wide breadth of a sector (Hryciw et al., 2023).

While this global discourse is vigorous, its empirical evidence base is skewed toward developed nations. There is a distinct lack of research on how AI is being adopted and perceived within the specific context of academic researchers and lecturers in higher education institutions (HEIs) in developing regions like Somalia. In such contexts, resource constraints, varying levels of digital literacy, and distinct disciplinary cultures may shape AI adoption and its ethical implications in unique ways. This study addresses this gap by focusing on academic researchers in Somali universities. It aims to understand not just if AI is used, but how the experience of using it (user experience) shapes perceptions of its integrity, and whether this relationship differs across academic disciplines. By testing these mediation and moderation mechanisms, the study provides novel, context-specific insights that move beyond descriptive accounts of AI use. The findings are intended to inform the development of relevant training and ethical frameworks for AI integration in Somali HEIs and similar settings, contributing a unique perspective from an under-researched region to the global literature.

Problem statement

The integration of Artificial Intelligence (AI) into academic research is accelerating within Somali higher education, driven by its potential to enhance productivity and innovation. However, this rapid uptake coincides with significant concerns among researchers regarding the ethical integrity and reliability of AI-generated outputs (e.g., plagiarism risks, data fabrication). This creates a tension: AI use is rising, yet trust in its integrity is not automatic. The mechanisms through which AI use translates into perceptions of integrity are unclear. Specifically, it is unknown whether a positive

User Experience (UX) acts as a crucial mediator in building trust, or if the relationship between AI use and perceived integrity is contingent upon the norms and practices of different Disciplinary Contexts (DC). Existing literature has largely examined these factors in isolation or within well-resourced, Western academic systems, neglecting the unique socio-technical environment of Somali universities. Consequently, there is no evidence-based guidance for Somali HEIs on how to foster responsible AI adoption. This study, therefore, investigates two core questions to address this gap: (1) To what extent does User Experience mediate the relationship between AI use and Perceived Integrity among Somali academic researchers? (2) Does Disciplinary Context moderate this relationship? By answering these, the study provides foundational evidence for developing tailored AI governance strategies that support research growth while safeguarding academic credibility in Somalia.

Value of the study

This study holds significant value across academic, practical, and policy dimensions. Academically, it advances the understanding of AI in research by exploring the understudied mediating role of user experience and moderating effect of disciplinary context on perceived integrity, particularly in Somalia's research context - filling crucial gaps in the existing literature. For practitioners like Somali researchers and academic administrators, the findings provide actionable insights for responsibly integrating AI tools to enhance research productivity while maintaining ethical standards, potentially informing training programs to improve AI literacy. At the policy level, the study offers evidence-based recommendations for developing context-specific AI governance frameworks and ethical guidelines in Somali higher education, supporting national efforts to strengthen research capacity and align with global academic standards.

Literature review

The integration of Artificial Intelligence (AI) into academic research has seen a significant rise, impacting various facets of scholarly communication. A critical evaluation reveals both facilitative benefits and ethical dilemmas concerning integrity.

AI benefits and use in research

AI tools are increasingly recognized for enhancing research efficiency and quality. They assist in systematic literature reviews, data analysis, and the drafting and editing of manuscripts, potentially leading to more robust outputs and clearer communication (Marmoa et al., 2024; Forgas et al., 2024). However, the effectiveness of these tools often depends on the availability of structured training, as a lack of formal training may result in misinformation and reduced content quality (Bhavsar et al., 2024; Ng et al., 2024). The perceived usefulness and ease of use of these technologies are key drivers of their adoption, as explained by technology acceptance models like UTAUT (Venkatesh, 2000).

Ethical risks and integrity concerns

Conversely, AI integration raises profound ethical questions. Issues of plagiarism, originality, bias reinforcement, and even the generation of fraudulent content threaten academic integrity (Amirjalili et al., 2024; Chen et al., 2024). While AI can promote critical thinking and collaboration, it also raises ethical challenges that require awareness and educational adaptations (Aisyah et al., 2024). While AI can improve writing and analysis, these advantages must be balanced against the necessity for careful evaluation of AI-generated content, as highlighted by ongoing discussions about authenticity and integrity (Kouam, 2024). This has spurred calls for clear institutional policies and ethical guidelines to govern AI use (Perkins and Roe, 2024; Ajwang and Ikoha, 2024). Trust in AI, therefore, is not solely based on performance but heavily on Perceived Integrity—the belief that the system operates ethically and transparently (Lalot and Bertram, 2025).

The role of user experience

The pathway from using AI to trusting it may depend heavily on UX. A system that is user-friendly, efficient, and reliable can foster positive attitudes and greater trust (Kim, 2024; Park et al., 2023). Conversely, poor UX, marked by inaccuracy or complexity, can undermine confidence and perceived integrity. Thus, UX is a plausible mediator in the relationship between AI use and integrity perceptions.

The influence of disciplinary context

Disciplinary norms and epistemic cultures significantly shape technology adoption. Fields with strong computational traditions (e.g., Engineering) may embrace AI more readily and trust its outputs differently compared to theory-heavy or interpretive fields (e.g., Humanities) (Liu et al., 2024; Fontaine et al., 2024). This suggests Disciplinary Context could moderate how AI use influences integrity perceptions.

The Somalia-specific evidence gap

A synthesis of the literature reveals a critical omission: nearly all empirical studies are situated in developed, well-resourced academic systems. There is a stark lack of investigation into how AI is adopted, experienced, and perceived in higher education contexts of developing nations like Somalia, where infrastructure, training, and research cultures present unique challenges and opportunities. This study directly addresses this gap.

Theoretical frameworks

This study is grounded in the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh, 2000) and trust-in-technology frameworks that emphasize Perceived Integrity as a key dimension (Lukyanenko et al., 2022). UTAUT posits that performance expectancy (usefulness) and effort expectancy (ease of use) drive

behavioral intention to use a technology, a notion supported by empirical studies on AI adoption in academia (Roy et al., 2024). In this context, “AI Use” represents the behavioral outcome. We extend this model by proposing that the consequences of use—specifically, the perception of the technology’s integrity—are not direct but are shaped by the user’s experience and their disciplinary environment.

AI use in academic research

Operationalized as the frequency and extent of applying AI tools to core research tasks such as literature discovery and synthesis (Granjeiro, 2025; Pinzolit, 2023), data analysis, and scholarly writing (Sharma et al., 2024).

User experience

refers to the pragmatic and hedonic qualities of interacting with an AI system. AI variants can tailor interaction to user needs, enhancing satisfaction (Luo et al., 2024; Kim et al., 2021). UX encompasses usability, efficiency, reliability, and overall satisfaction (Kim, 2024; Dave et al., 2023; Wang et al., 2023), and is crucial for user comfort and continued engagement (Park et al., 2023; Wei, 2024). Acceptance depends on understanding user factors (Jiang et al., 2024), including perceptions of effectiveness and likability (Zheng and Wang, 2022). We hypothesize that a positive UX translates the functional use of AI into a trusting belief in its integrity.

Perceived integrity

Generally refers to the extent to which users believe that an AI system adheres to ethical principles of honesty, fairness, transparency, and accountability in its operations and outputs (Lalot and Bertram, 2025). It is a core antecedent of trust. Research indicates that positive attitudes toward AI tools, shaped by integrity perceptions, promote acceptance (Al-Bukhrani et al., 2025).

Disciplinary context

Encompasses the distinct norms, methodologies, and epistemic values of an academic field that shape how knowledge is produced and validated. Disciplinary differences significantly determine AI acceptance and integration (Liu et al., 2024), with fields varying in their epistemic readiness and evaluation biases (Seeber et al., 2022; Fontaine et al., 2024; Fecher and Hebing, 2021). We theorize it as a boundary condition that may strengthen or weaken the AI Use-Perceived Integrity relationship.

Conceptual model and hypotheses

The conceptual model (Figure 1) illustrates these relationships. It posits a direct positive effect of AI Use on Perceived Integrity (H1). More importantly, it proposes that this effect is transmitted through User Experience (H2: Mediation). Furthermore, it suggests that the strength of the direct relationship is conditional upon the Disciplinary Context (H3: Moderation). This model allows us to test not just if AI use relates to integrity, but how and under what conditions.

Based on theoretical framework, the study tests the following hypotheses:

H1: AI use in academic research has a statistically significant positive effect on perceived integrity.

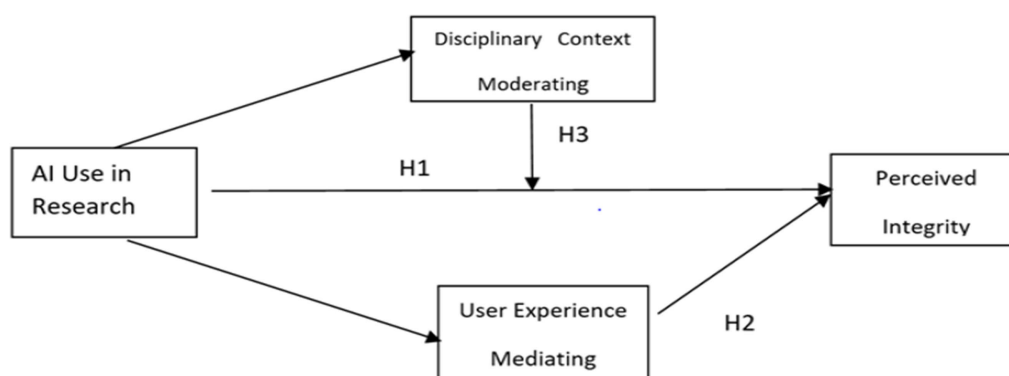


FIGURE 1
Conceptual research model.

H2: User experience mediates the relationship between AI use in academic research and perceived integrity.

H3: Disciplinary context moderates the relationship between AI use in academic research and perceived integrity.

Method

Research design

A quantitative, cross-sectional research design was employed to examine the hypothesized relationships.

Study setting and population

The study was conducted among academic researchers (including lecturers and postgraduate students actively engaged in research) from three major universities in Mogadishu, Somalia: Mogadishu University, Somali National University and SIMAD University. These institutions were selected due to their established postgraduate programs and research activities, providing a relevant and accessible population. The total target population of active academic researchers across these institutions was estimated to be approximately 250.

Sampling and recruitment

A random sampling strategy was used. The sample size was determined *a priori* using G*Power 3.1 for a linear multiple regression (fixed model, R^2 increase) with an effect size $f^2 = 0.35$, $\alpha = 0.05$, and power = 0.95, which suggested a minimum of 50 participants (Kang, 2021). To account for potential non-response and ensure robustness, we aimed for a larger sample. Eligibility criteria included: (1) being an active lecturer or postgraduate researcher at one of the target universities, and (2) having some level of awareness or experience with AI tools for research. Recruitment was conducted over 4 weeks through email lists and coordination with departmental heads. Of 150 invitations distributed, 100 completed responses were received (response rate = 66.7%). Non-response was primarily due to lack of time or no experience with AI.

Instrument and measures

Data were collected via a structured online questionnaire. All constructs were measured using adopted scales from prior studies on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The scales measured: AI Use in Research (AIU, 5 items, e.g., “I frequently use AI in different parts of my research”) (Al-Rousan et al., 2025); User Experience (UX, 5 items, e.g., “The AI system has a user-friendly interface”) (Yin and Li, 2024); Perceived Integrity (PI, 5 items, e.g., “I am confident that AI integration enhances research rigor”; Oyekunle, 2024); and Disciplinary Context (DC, 4 items, e.g., “I believe that the use of AI technologies enhances the learning experience within my academic discipline”) (Santos et al., 2024). The instrument was first piloted with 10 researchers to check for clarity and contextual relevance; minor wording adjustments were made based on feedback; the following Table 1 is Construct Measurement Summary.

Data analysis plan

Data analysis was conducted using SPSS version 24, which utilized descriptive statistics (weighted means and standard deviations) on a survey Likert data scale, and the mediating and moderating analyses were examined through JAMOV version 2.6.44.

Ethical considerations

Participation was voluntary and anonymous. Informed consent was obtained electronically prior to starting the questionnaire. Data were stored securely on a password-protected computer and will be deleted after 5 years.

Demographic profile of respondents

Table 2 presents the demographic characteristics of the 100 respondents. The sample was predominantly male (88%), reflecting a known gender disparity in higher education roles in the region (Ahmed et al., 2023). The majority were aged 36–40 years (32%) and were Master’s level researchers (65%). Respondents came primarily from Social Sciences & Humanities (35%) and Health Sciences/Medicine (31%).

TABLE 1 Construct measurement summary.

Construct	No. of Items	Cronbach's α
AI use	5	0.8
User experience	5	0.7
Perceived integrity	5	0.7
Disciplinary context	4	0.77

Results

Table 3 depicts Harman's Single Factor Test results showing that the first component explains 38.564% of the total variance, which is below the 50% threshold. This suggests that common method bias (CMB) is not a significant concern in the data, as no single dominant factor accounts for the majority of variance.

Table 4 presents the mean scores and standard deviations of survey items measuring four key constructs related to AI use in academic research:

AI use in academic research

The findings indicate a high level of AI integration into research practices among Somali academics (Grand Mean = 3.76). Respondents reported frequent use of AI across various research stages and expressed confidence in selecting appropriate tools. Notably, there is a strong commitment to professional development related to AI (mean = 3.91), reflecting a proactive stance toward adopting and mastering emerging technologies.

User experience

Overall, user experience with AI systems was highly positive (Grand Mean = 3.77). Participants found AI interfaces user-friendly and easy to learn, and they valued the efficiency gains in their workflows. However, a notable exception was the moderate rating for the accuracy and reliability of AI outcomes (mean = 3.31), signaling a specific area of concern that may affect trust.

Perceived integrity

Participants generally perceived AI as enhancing research integrity (Grand Mean = 3.58). They agreed that AI benefits outweigh integrity risks and emphasized the need for continuous policy monitoring. Nevertheless, ratings regarding the ethical integrity of AI systems themselves were more moderate (mean = 3.31 on relevant items), suggesting nuanced trust.

Disciplinary context

AI was viewed as well-aligned with disciplinary norms and innovative practices across fields (Grand Mean = 3.76). Respondents believed AI enhances learning and problem-solving within their disciplines, indicating broad contextual acceptance.

TABLE 2 Demographic profile of respondents ($N = 100$).

Characteristic	Category	Frequency	Percentage
Gender	Male	88	88%
	Female	12	12%
Age group	26–30 years	15	15%
	31–35 years	25	25%
	36–40 years	32	32%
	41–45 years	20	20%
	46 + years	8	8%
Highest qualification	Bachelor's	13	13%
	Master's	65	65%
	PhD	22	22%
Disciplinary field	Social sciences and humanities	35	35%
	Health sciences/medicine	31	31%
	Business and economics	21	21%
	Engineering and technology	11	11%
	Natural sciences	2	2%

Figure 2 presents a comparison of the grand means for four study factors: AI Use in Academic Research, User Experience, Perceived Integrity, and Disciplinary Context. All means relatively high, indicating positive responses across all factors, with User Experience having the highest mean. Standard deviations are smaller, suggesting less variability in responses, although Perceived Integrity shows slightly more variation than the other factors. Overall, the figure suggests that participants generally have positive perceptions of AI use, user experience, integrity, and disciplinary context in academic research.

Table 5 presents the correlation matrix analyzes the relationships between four key variables in AI adoption in academic research: AI Use in Academic Research, User Experience, Perceived Integrity, and Disciplinary Context. With a sample of 100 respondents, the study finds significant positive correlations at the 0.01 level. Notably, AI Use correlates strongly with User Experience ($r = 0.658$), Perceived Integrity ($r = 0.615$), and Disciplinary Context ($r = 0.628$). The strongest links are between User Experience and both Perceived Integrity ($r = 0.678$) and Disciplinary Context ($r = 0.674$), indicating that positive AI experiences boost trust and alignment with supportive disciplines. Additionally, Perceived Integrity correlates with Disciplinary Context ($r = 0.607$), emphasizing the connection between ethical perceptions and field norms. All correlations suggest moderately strong relationships without multicollinearity issues. Based on these findings, we confirm Hypothesis 1, which stated that AI use in academic research has a statistically significant positive effect on perceived integrity ($p = 0.000$) less than ($\alpha = 0.05$). Therefore, we reject the null hypothesis and accept the alternative hypothesis.

TABLE 3 Result of Harman's single factor test for common method bias (CMB).

Total variance explained						
Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
AIUR1	7.327	38.564	38.564	7.327	38.564	38.564
AIUR2	1.452	7.643	46.207			
AIUR3	1.282	6.749	52.956			
AIUR4	1.061	5.586	58.542			
AIUR5	0.954	5.019	63.562			
UX1	0.829	4.364	67.926			
UX2	0.771	4.059	71.985			
UX3	0.668	3.514	75.498			
UX4	0.636	3.347	78.845			
UX5	0.600	3.159	82.004			
PI1	0.564	2.969	84.974			
PI2	0.511	2.689	87.662			
PI3	0.435	2.289	89.951			
PI4	0.416	2.192	92.143			
PI5	0.400	2.106	94.250			
DC1	0.362	1.903	96.153			
DC2	0.308	1.620	97.773			
DC3	0.244	1.282	99.054			
DC4	0.180	0.946	100.000			

Extraction method: Principal component analysis.

Table 6 depicts the mediation analysis revealing important insights into the relationships among AI use, user experience, and perceived integrity. The indirect effect of AI use on perceived integrity through user experience is estimated at 0.280, with a standard error of 0.0628, yielding a Z-value of 4.47 and a highly significant p -value ($p < 0.001$). This suggests that user experience significantly mediates the relationship between AI use and perceived integrity. The direct effect of AI use on perceived integrity is also significant, with an estimate of 0.262, a standard error of 0.0820, and a Z-value of 3.20 ($p = 0.001$), indicating that AI use directly influences perceived integrity as well. The total effect, combining both direct and indirect pathways, is 0.542, with a standard error of 0.0695 and a Z-value of 7.80 ($p < 0.001$), confirming the overall significance of the model. Thus, Hypothesis 2, which stated that User experience mediates the relationship between AI use in academic research and perceived integrity was supported ($p < 0.001$).

Table 7 displays the results of the moderation analysis, illustrating how disciplinary context affects the relationship between AI use and perceived integrity. The estimate for AI use is 0.33694, with a standard error of 0.0749 and a Z-value of 4.496, indicating a strong positive effect on perceived integrity that is statistically significant ($p < 0.001$). This suggests that higher AI use is associated with greater perceived integrity. The estimate for disciplinary context is even higher at 0.45606, also statistically significant ($p < 0.001$), indicating a robust positive influence on perceived integrity. However, the interaction term for AI use and disciplinary context has an estimate of -0.00294 , a standard error of 0.0192, and a Z-value of -0.153 , with a p -value of 0.878, suggesting no significant moderation effect. Thus, Hypothesis 3, which stated that

disciplinary context moderates the relationship between AI use in academic research and perceived integrity, was not supported ($p = 0.878$).

Exploration of non-significant moderation

Given the non-significant interaction effect, we explored potential explanations. First, the measure of Disciplinary Context focused on perceived alignment and innovativeness of AI within a field, which may be uniformly high across disciplines in our sample, reducing variability. Second, the benefits of AI for research integrity (e.g., efficiency, rigor) may be perceived similarly regardless of discipline among Somali researchers who are in earlier stages of AI adoption. Third, the sample distribution across disciplines (e.g., fewer from Natural Sciences) may have limited our ability to detect moderation. These possibilities suggest that while disciplinary norms exist, their moderating role on the *specific relationship* between AI use and integrity perceptions may be less pronounced in this context.

Discussion of findings

AI use in academic research

The findings indicate that researchers are actively incorporating AI into their workflows, viewing it as valuable for enhancing literature

TABLE 4 Results for the study factors' items.

Items	Study variables			
	AI use in academic research			
	Statement	Mean	Std. Deviation	Decision
AIUR1	I frequently use AI in different parts of my research.	3.77	0.908	High
AIUR2	Using AI helps me do better literature reviews, and analyze data.	3.74	0.939	High
AIUR3	I confidently choose AI tools to enhance my research and results.	3.66	0.977	High
AIUR4	AI fosters creativity in my research by providing innovative problem-solving methods.	3.73	0.941	High
AIUR5	I pursue professional development to stay updated on emerging AI technologies for academic research	3.91	0.793	High
	Grand mean	3.76	0.91	High

	User experience	Mean	Std. Deviation	Decision
UX1	The AI system has a user-friendly interface for easy navigation.	3.82	0.914	High
UX2	I quickly grasp how to use the AI system, ensuring smooth learning.	3.91	0.830	High
UX3	The AI system completes tasks quickly, enhancing my workflow.	4.00	0.921	High
UX4	The AI system produces accurate and reliable outcomes for my research.	3.31	1.061	Moderate
UX5	My experience with the AI system has been positive, meeting my performance expectations	3.82	0.903	High
	Grand mean	3.77	0.92	High

	Perceived integrity	Mean	Std. Deviation	Decision
PI1	I am confident that AI integration enhances research rigor and credibility while upholding ethical standards.	3.56	0.925	High
PI2	AI benefits, like efficiency and advancement, outweigh/exceed risks to research integrity.	3.54	0.947	High
PI3	Continuous monitoring and improvement of AI policies are essential for ethical research practices.	3.93	0.832	High
PI4	The AI system operates with high ethical integrity and adheres to established principles.	3.62	0.919	Moderate
PI5	The AI system is accountable for its actions and decisions.	3.25	1.104	High
	Grand mean	3.58	0.94	High

	Disciplinary context	Mean	Std. Deviation	Decision
DC1	I believe that the use of AI technologies enhances the learning experience within my academic discipline.	3.81	0.825	High
DC2	Integrating AI tools in my field provides innovative approaches to problem solving.	3.76	0.830	High
DC3	The integration of AI tools is consistent with the current trends and practices in my academic discipline.	3.84	0.735	High
DC4	AI applications in my field align with its professional and disciplinary standards.	3.64	0.811	High
	Grand Mean	3.76	0.80	High

Scale: Very High to Very Low: Very High (4.20–5.00), High (3.40–4.19), Moderate (2.60–3.39), Low (1.80–2.59), and Very Low (1.00–1.79).

reviews, data analysis, and fostering creativity. The strong commitment to professional development suggests a proactive approach to adopting these technologies, aligning with the performance expectancy dimension of UTAUT. This widespread adoption reflects AI's growing integral role in the academic research landscape, consistent with prior studies on the transformative impact of AI tools (Ng et al., 2024; Kouam, 2024).

User experience

The generally positive user experience, characterized by user-friendly interfaces and efficient workflows, supports the effort expectancy dimension of UTAUT. However, concerns regarding the reliability and accuracy of AI outcomes highlight a critical area for improvement, as trust in AI systems is heavily influenced by perceived

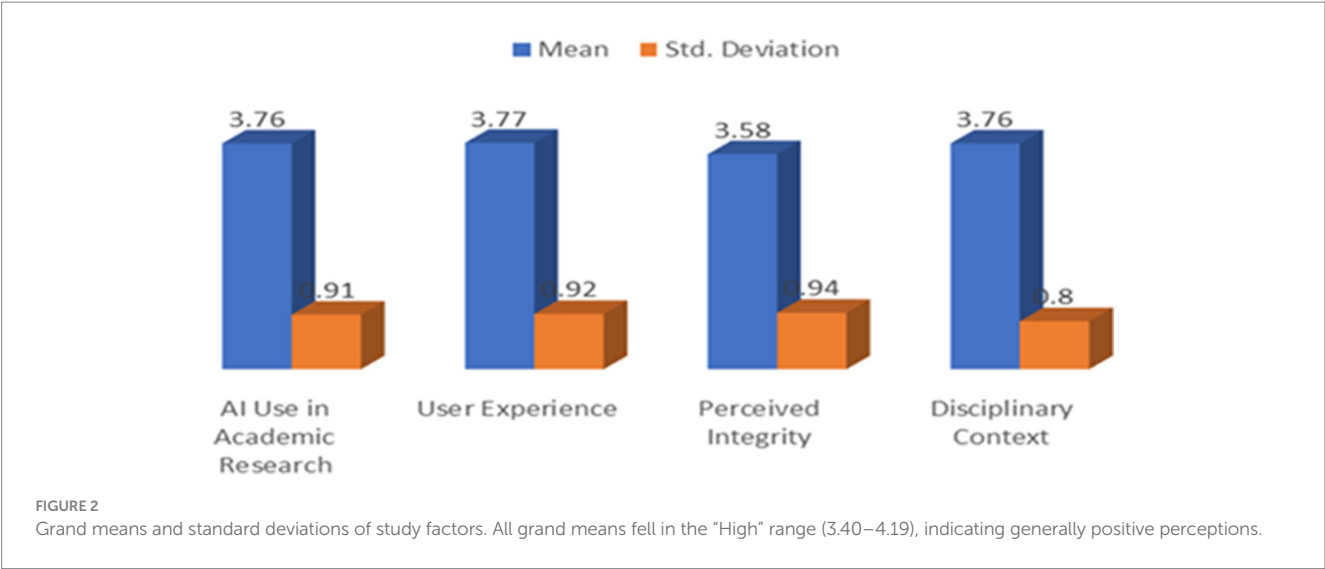


TABLE 5 Correlations matrix.

Variables		AI use in academic research	User experience	Perceived integrity	Disciplinary context
AI Use in academic research	Pearson Correlation	1	0.658**	0.615**	0.628**
	Sig. (2-tailed)		0.000	0.000	0.000
	N	100	100	100	100
User experience	Pearson Correlation	0.658**	1	0.678**	0.674**
	Sig. (2-tailed)	0.000		0.000	0.000
	N	100	100	100	100
Perceived integrity	Pearson Correlation	0.615**	0.678**	1	0.607**
	Sig. (2-tailed)	0.000	0.000		0.000
	N	100	100	100	100
Disciplinary context	Pearson Correlation	0.628**	0.674**	0.607**	1
	Sig. (2-tailed)	0.000	0.000	0.000	
	N	100	100	100	100

**Correlation is significant at the 0.01 level (2-tailed).

TABLE 6 Mediation estimates.

Effect	Estimate	SE	Z	p
Indirect	0.280	0.0628	4.47	<0.001
Direct	0.262	0.0820	3.20	0.001
Total	0.542	0.0695	7.80	<0.001

*Dependent variable: perceived integrity.

reliability (Nasra et al., 2024). This mixed UX—positive in usability but cautious in trust—underscores the nuanced role UX plays in mediating the relationship between use and integrity perceptions.

Perceived integrity

Participants generally believe AI can enhance research rigor and credibility while upholding ethical standards, and they emphasize the

need for continuous policy monitoring. However, moderate scores on ethical integrity indicate lingering concerns. This aligns with discussions on the necessity of legal, regulatory, and ethical frameworks to ensure accountability and uphold integrity in AI-assisted research (O’Sullivan et al., 2019). The belief in AI’s accountability reinforces trust but points to the need for further efforts in ethical safeguards.

Disciplinary context

Researchers perceive AI as enhancing learning and problem-solving in their disciplines, consistent with current trends and standards. This suggests AI is well-integrated across various academic contexts, supporting its role as an adaptive and innovative educational tool (Abbas et al., 2023; Guerrero et al., 2020). However, the non-significant moderating effect of disciplinary context suggests that,

TABLE 7 Moderation estimates.

	Estimate	SE	Z	p
AI use in academic research	0.33694	0.0749	4.496	<0.001
Disciplinary context	0.45606	0.0932	4.895	<0.001
AI use in academic research * disciplinary context	−0.00294	0.0192	−0.153	0.878

*Dependent variable: perceived integrity.

in this setting, the perceived benefits and integrity of AI may transcend disciplinary boundaries.

theories, especially in developing academic systems where trust is critical.

Moderation analysis: the non-significant role of discipline

The moderation analysis found no significant interaction between AI use and disciplinary context on perceived integrity. This suggests that the positive effect of AI use on integrity perceptions is consistent across different fields in the Somali academic context. This finding contrasts with literature emphasizing strong disciplinary differences in technology adoption (Liu et al., 2024; Fontaine et al., 2024) but aligns with perspectives that core trust drivers like perceived integrity can be universal across contexts (Lalot and Bertram, 2025; Shinnners et al., 2019). Possible explanations include the relatively nascent stage of AI adoption in Somalia, where researchers across disciplines share similar challenges and expectations, or the measurement of disciplinary context focusing on general alignment rather than deep epistemic differences.

Mediation analysis: the central role of user experience

The mediation analysis confirms that user experience significantly mediates the relationship between AI use and perceived integrity. The significant indirect effect underscores that a positive UX enhances trust in AI's integrity, while the remaining direct effect suggests other unmeasured factors also contribute. This aligns with technology acceptance research where perceived usefulness and ease of use shape attitudes and trust (Liang et al., 2019; Venkatesh, 2000).

Theoretical implications

This study adds to two important theories: The Unified Theory of Acceptance and Use of Technology (UTAUT) and trust-based models of technology. First, while UTAUT usually focuses on why people start using technology, we show that actually using AI can shape how much people trust its integrity afterward. Second, we found that user experience (UX) is the key link explaining this connection—how easy and useful AI feels directly affects whether researchers see it as ethical. Third, contrary to expectations, disciplinary differences did not change this relationship in our Somali sample. This suggests that in settings where AI is still new and resources are limited, practical and ethical concerns may matter more than field-specific norms. Together, these insights call for a more context-aware approach to technology adoption

Contribution and novelty

This study contributes novel insights by providing empirical evidence from a Somali academic context, where AI research is underexplored. It uniquely tests dual mechanisms (mediation and moderation) in the AI-integrity relationship, revealing the paramount importance of user experience over disciplinary boundaries in this setting. These findings challenge the assumed primacy of disciplinary culture in technology adoption literature and underscore the need for context-specific models of AI integration and governance in resource-constrained higher education environments.

Conclusion

This study aimed to examine how AI use in academic research is associated with perceived integrity, and to test whether user experience mediates and disciplinary context moderates this relationship within Somali higher education. The findings reveal a positive trend in AI adoption, with researchers integrating AI tools to enhance literature reviews, data analysis, and research outcomes, and demonstrating commitment to related professional development. Overall, perceived integrity was high, suggesting that researchers believe AI can enhance research rigor and ethical standards, though concerns about reliability and ethical integrity persist. Key synthesized takeaways are as follows: First, user experience plays a significant mediating role, indicating that a positive experience with AI systems strengthens perceptions of their integrity. Second, disciplinary context did not moderate the relationship, implying that the association between AI use and perceived integrity is consistent across different academic fields in this context. These insights advance our understanding of the mechanisms linking AI adoption to trust in an under-researched setting. The study contributes by providing evidence-based insights from Somalia, highlighting the central role of user experience and the context-dependent nature of disciplinary influences. However, limitations should be noted, including the cross-sectional design, which precludes causal inferences; the gender imbalance in the sample, which may affect generalizability; and the focus on a single national context. Future research should employ longitudinal designs, include more diverse samples, and explore additional potential moderators such as institutional support or researcher seniority. Such work will further inform the development of responsible AI integration policies tailored to the needs of developing higher education systems.

Recommendations

Based on the findings of this study, the following actionable recommendations are prioritized for academic institutions and researchers in Somalia and similar contexts:

- Develop and implement structured AI literacy training programs focused not only on tool usage but also on critical evaluation of AI outputs, ethical guidelines, and integrity verification. This addresses the UX-mediating role and reliability concerns identified in the study.
- Establish clear, context-appropriate disclosure guidelines for AI use in research proposals, manuscripts, and theses to promote transparency and accountability, thereby bolstering perceived integrity.
- Integrate mandatory AI-output verification and integrity checks into the research workflow and supervision processes to mitigate risks related to inaccuracy, plagiarism, and data fabrication.
- Form institutional AI ethics committees with clear Terms of Reference (TOR) to develop, review, and regularly update AI use policies, provide ethical oversight, and handle related grievances.
- Create equitable access plans to ensure AI tools and necessary computational resources are available across disciplines and to all researchers, preventing a digital divide that could exacerbate existing inequalities.
- Define and monitor key evaluation metrics for AI integration, such as user satisfaction scores, perceived integrity measures, and research output quality indicators, to iteratively improve support systems and policies.
- For researchers, it is recommended to: engage proactively with available training, apply critical scrutiny to AI-generated content, maintain transparent records of AI use, and participate in interdisciplinary dialogs to shape locally relevant best practices.

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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